

# Qatar-1b

R-2732

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_Qatar-1\_110820 · created 2026-09-02T00:29:16+00:00

BENCHMARK: ACCURATE — fit consistent with published values

## Results

|                                                   |                                  |
|---------------------------------------------------|----------------------------------|
| Mid-Transit Time (Tmid)                           | 2455793.89605 +/- 0.0004 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.144 +/- 0.043                  |
| Transit depth (Rp/Rs)^2                           | 2.06 +/- 1.22 [%]                |
| Orbital Inclination (inc)                         | 82.2 +/- 2.54                    |
| Airmass coefficient 1 (a1)                        | 0.928 +/- 0.018                  |
| Airmass coefficient 2 (a2)                        | 0.041 +/- 0.022                  |
| Scatter in the residuals of the lightcurve fit is | 1.95 %                           |
| Best Comparison Star                              | #2 - [521, 237]                  |
| Optimal Aperture                                  | 3.01                             |
| Optimal Annulus                                   | 5.2                              |
| Transit Duration (day)                            | 0.049 +/- 0.024                  |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.144 ± 0.043 vs 0.1463 (deviation 0.05σ) |
| Duration fit vs published | 0.049 d vs 0.0692 d (deviation 0.77σ)     |
| Mid-time O–C              | 0.7 min                                   |
| Depth SNR                 | 3.1                                       |
| Residual scatter          | 1.95 %                                    |

## Light curve

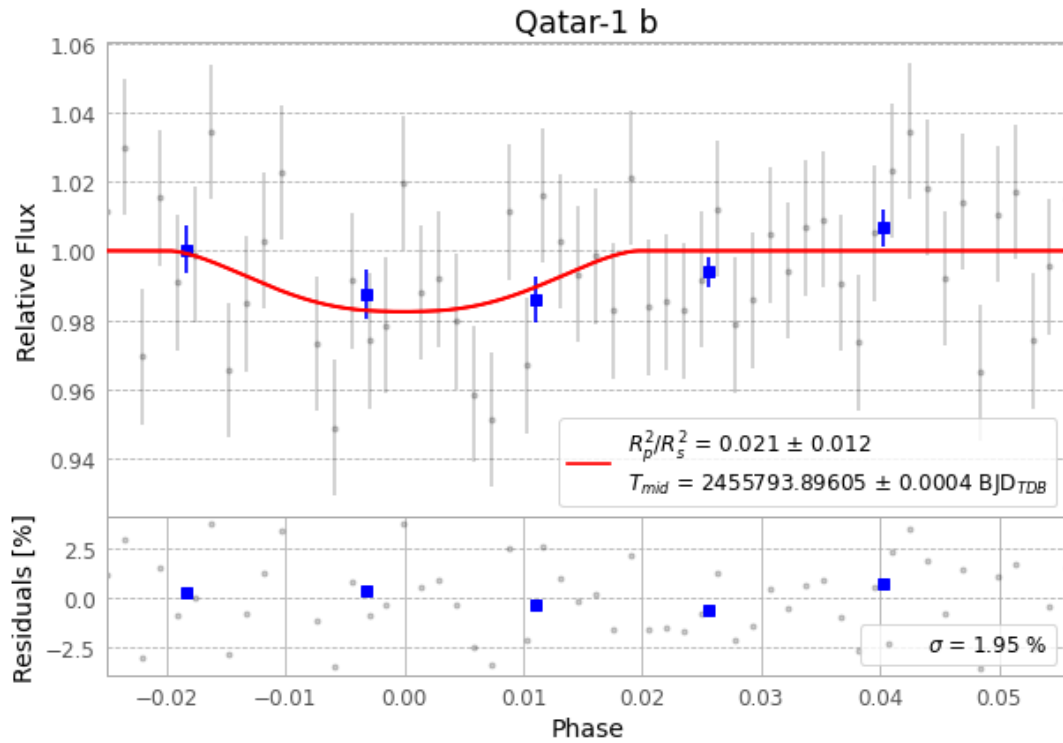


Figure 1 — FinalLightCurve\_Qatar-1 b\_2011-08-20.png (SHA-256 21361c001022dace...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [476, 294], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 b01bca39e30abdd9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit a4253ce

## Data provenance

56 FITS frames (37 MB), Qatar-1110820083612.FITS → Qatar-1110820112113.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-110820.log (8e120e281bed...), FinalLightCurve\_Qatar-1 b\_2011-08-20.png (21361c001022...), FinalLightCurve\_Qatar-1 b\_2011-08-20.csv (b149f7bea318...)

## Compute resources

Wall time 136.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-02 00:29Z.